# Symbolic vs Numeric AI

Numeric/Statistical AI is what we have seen before, like chatgpt that gives you a statistical outcome.

Symbolic AI tries to infer new proofs.

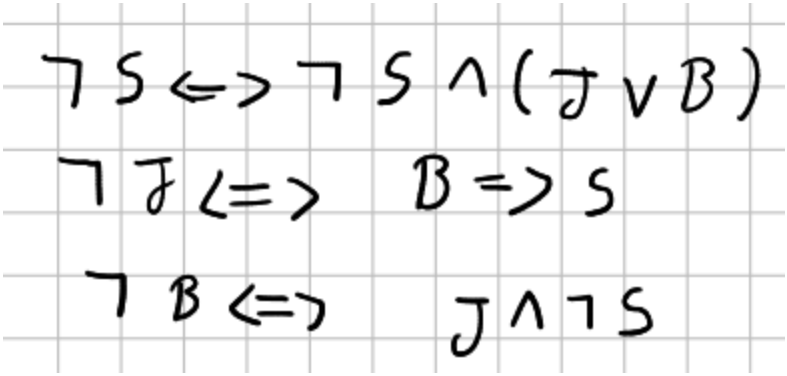
# What is logic?

### slide 15

1. Bernard => guilty, Bernard => Sophie, Sophie => guilty
2. Jean => guilty, Jean => not Sophie, Sophie => not guilty
3. Sophie => not guilty, Bernard or Jean => guilty

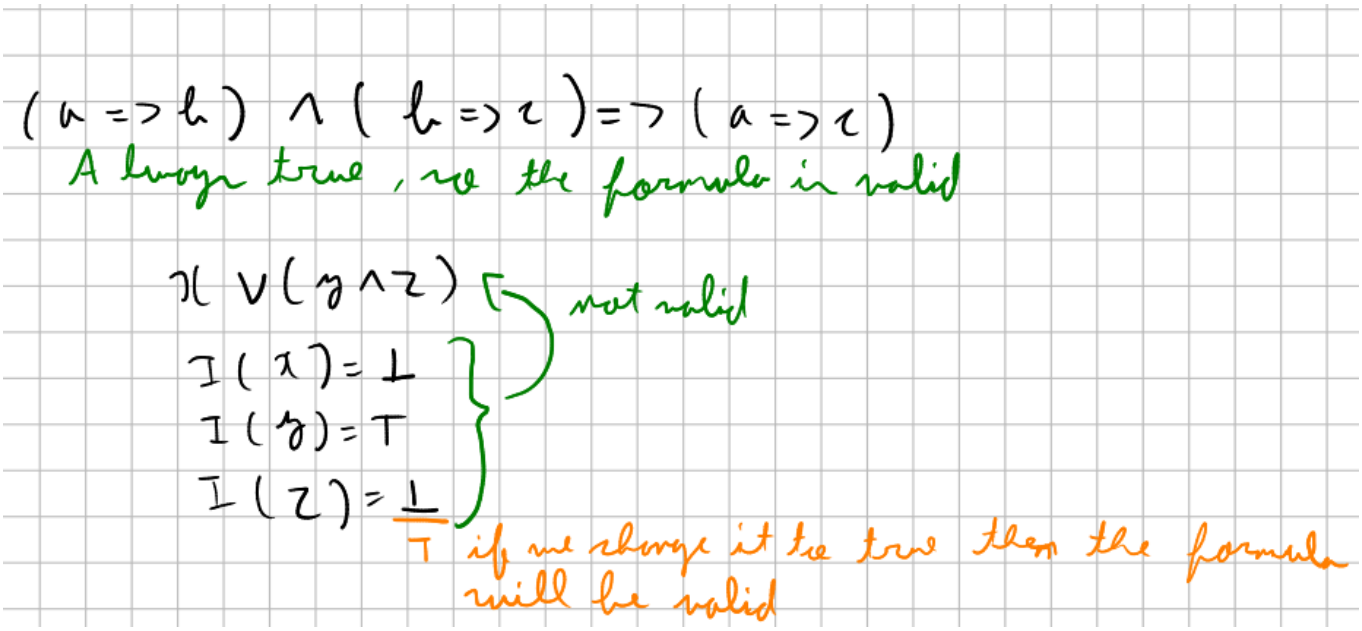
Correction

B (Bernard guilty)
S (Sophie guilty)
J (Jean guilty)

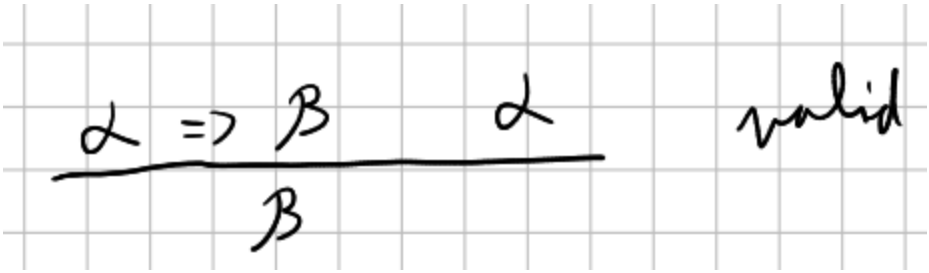



# Semantic models

A formula is valid if the interpretation of the this model is true

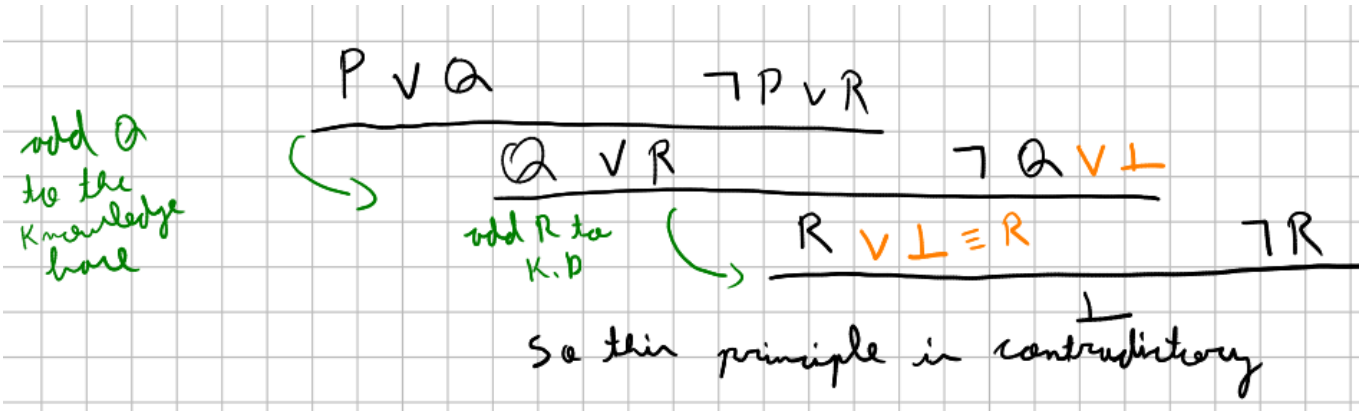



## Valid syntactic rules?




If alpha is true then beta is true so the proposition is valid. The proposition (rule) on top should imply the below one.

## Resolution Principle



$\Gamma = \{ 1.\ P \vee Q,\ 2.\ \neg P \vee R,\ 3.\ \neg R \vee S,\ 4.\ \neg S \vee T,\ 5.\ \neg T,\ 6.\ \neg Q$

so $\Gamma' = \Gamma \setminus \{\neg Q\}$

$\Gamma' \models Q$ implies

$P \vee Q \quad \neg P \vee R$

$Q \vee R \quad \neg R \vee S$

$Q \vee S \quad T \vee \neg S$

$T \vee Q \quad \neg T$

$\neg Q \quad Q$

$\bot$

contradictory

**Conjunctive normal form:**

$(P \vee Q) \wedge (\neg P \vee R) \wedge \ldots$

$P \Rightarrow Q \rightarrow \neg P \vee Q$

$P \Leftrightarrow Q$

$(\neg P \vee Q) \wedge (\neg Q \vee P)$

$\neg S \Leftrightarrow \neg S \wedge (J \vee B)$

$\neg J \Leftrightarrow B \Rightarrow S$

$\neg B \Leftrightarrow J \wedge \neg S$

$CNF \leftarrow (l_{11} \vee \ldots l_{1k1}) \wedge (l_{21} \vee \ldots \vee l_{2k2}) \wedge \ldots$

Bernard, Sophie, Jean example

$(\neg S \Leftrightarrow \neg S \wedge (J \vee B)) \wedge (\neg (\neg J \Leftrightarrow B \Rightarrow S)) \wedge (\neg (B \Leftrightarrow J \wedge \neg S))$

simple example $\neg S \Leftrightarrow (\neg S \wedge (B \vee J))$

$\neg S \Rightarrow (\neg S \wedge (B \vee J)) \wedge (\neg S \wedge (B \vee J) \Rightarrow \neg S)$

$(S \vee (\neg S \wedge (B \vee J))) \wedge ((S \vee (\neg B \wedge \neg J)) \vee \neg S)$

$(S \vee \neg S) \wedge (S \vee B \vee J) \wedge ((S \vee \neg B) \wedge (S \vee \neg J)) \vee \neg S$

$\wedge (S \vee \neg S \vee (\neg B \wedge S \wedge S)) \wedge \neg S \vee S \vee \neg J$